Sensorless Drive Diagnosis

ML Assignment 2. Verified local backup prepared 2026-11-08 00:53.

1. GitHub repository link

https://github.com/2026ak05140-dev/MLAssignment2

The public repository is accessible, but its published README differs from this local revision. Publish the current source, test CSV, fitted models, and README before submission.

2. Live Streamlit app link

https://mlassignment2-nz97nufqwetuqsaxhdjt6c.streamlit.app/

Deployment verification is incomplete. This environment redirects the URL to a network Site Blocked page; a separate fetch returned HTTP 403. This does not prove the app itself is unavailable. Redeploy and verify the interactive frontend from an allowed network.

3. BITS Virtual Lab execution evidence

> BITS Virtual Lab evidence is unavailable. Replace this paragraph and the following supporting local evidence with one genuine BITS execution screenshot after a successful lab run. Local screenshots do not fulfill the rubric's 1-mark lab requirement.

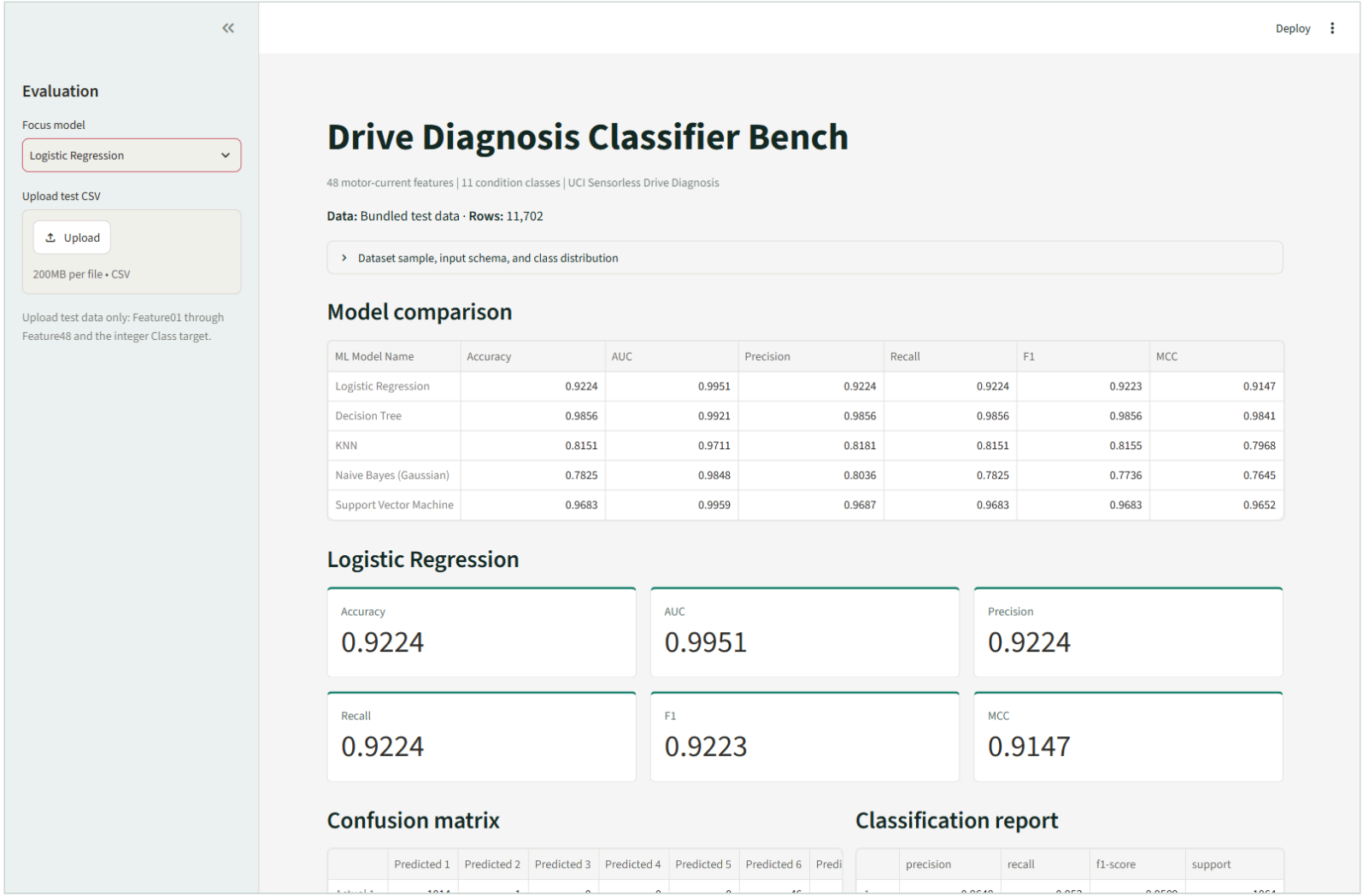
Local checks on 2026-11-08: the final six-test regression run passed in approximately 161 seconds. The full 11,702-row app comparison reproduces all 30 stored metrics within 1e-12. Lint, formatting, and compilation checks passed. All five model selections render six metrics and an 11-class report. The browser accepts the test CSV and rejects missing features. The experiment uses 58,509 source records, 48 inputs, and 11 classes without subsampling.

Word is an editing copy. The assignment requires one final PDF in this section order. Recheck links and layout after edits. Local validation does not certify originality or public deployment; review the assignment's AI-use rules and understand the implementation.

Supporting local evidence 1

Five-model comparison and six Logistic Regression metrics on 11,702 test records.

Captured locally on 2026-11-08. Not BITS Virtual Lab evidence.

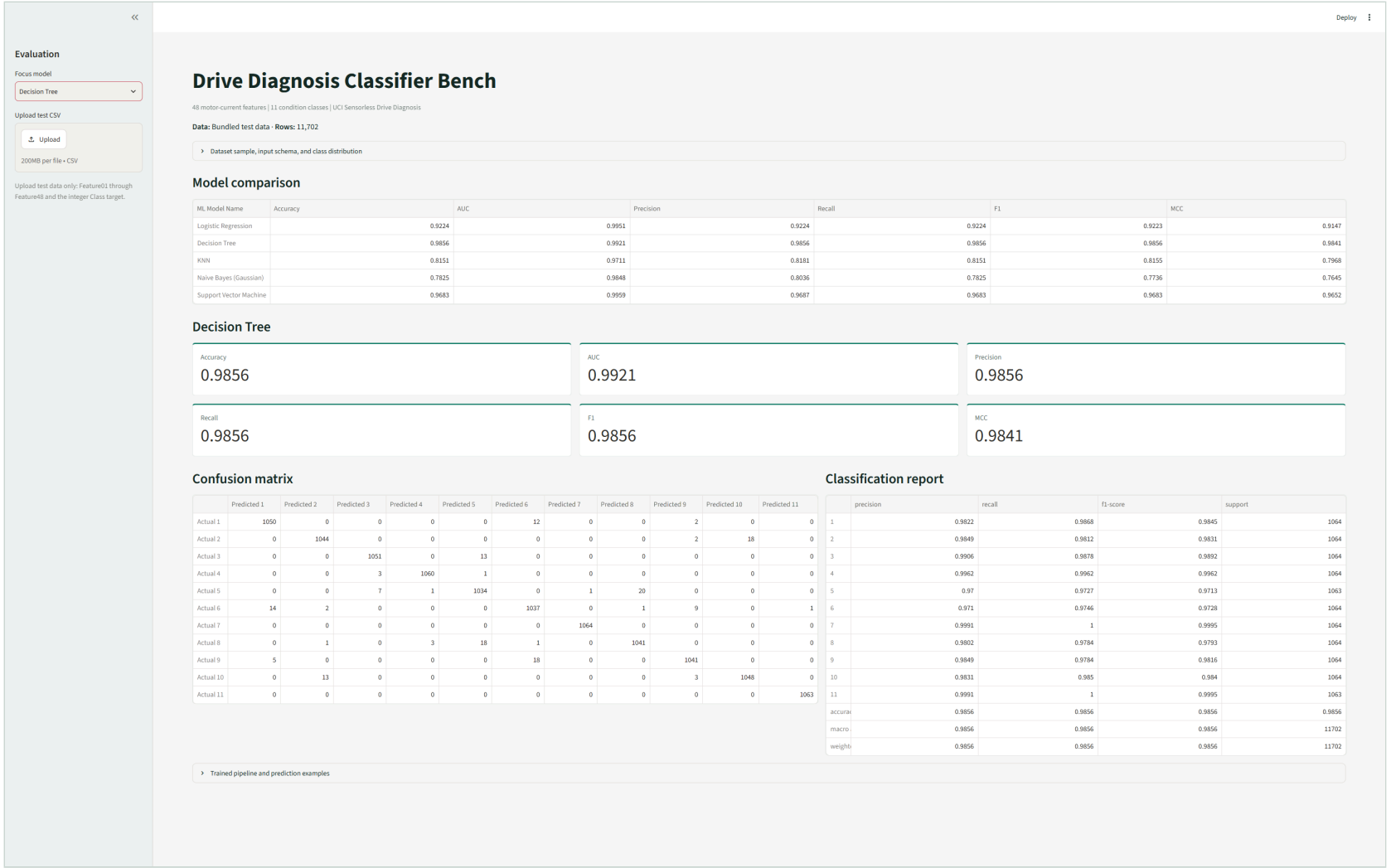


Five-model comparison and six Logistic Regression metrics on 11,702 test records.

Supporting local evidence 2

Winning Decision Tree: six metrics, all 11 confusion-matrix rows, and classification report.

Captured locally on 2026-11-08. Not BITS Virtual Lab evidence.

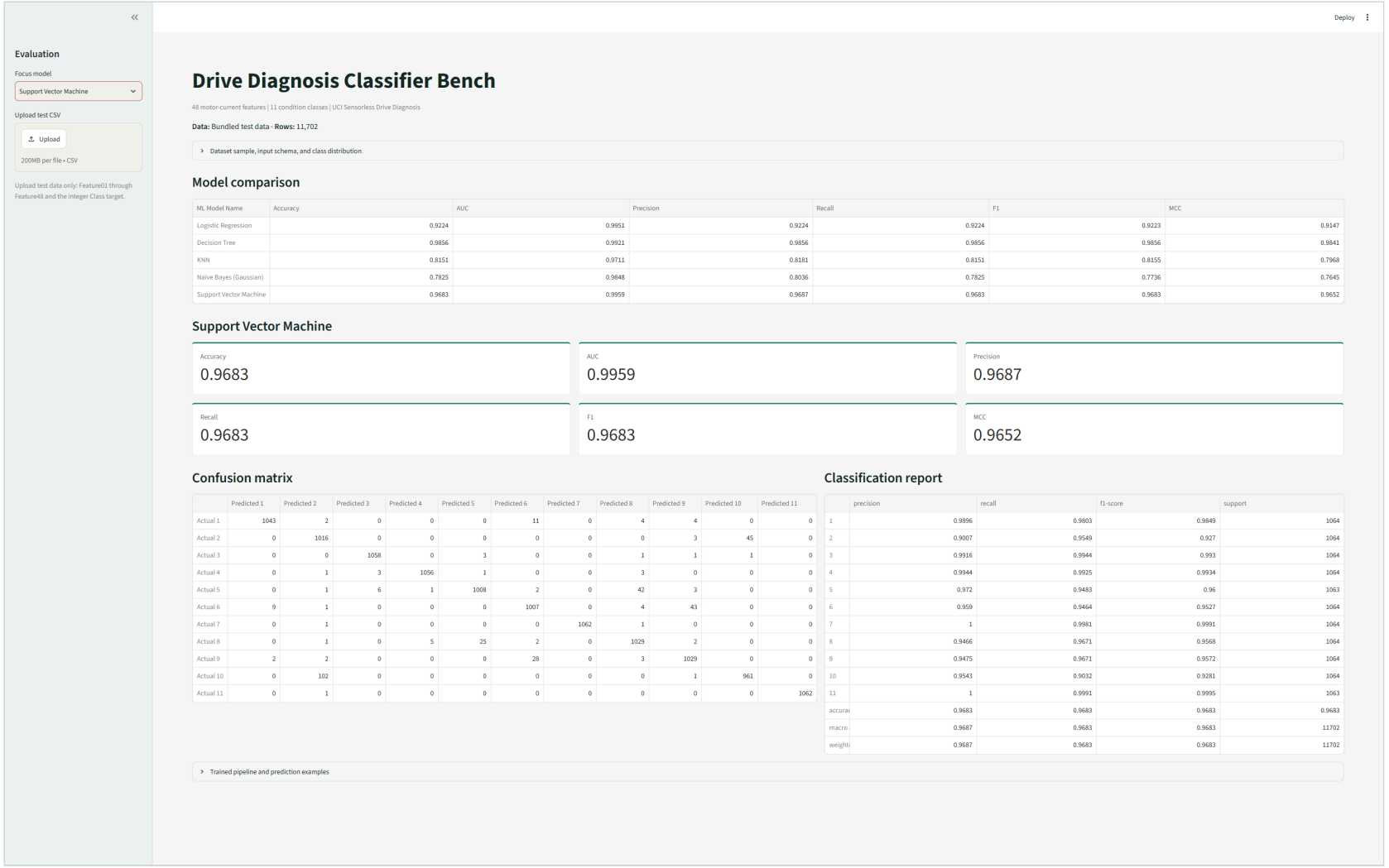


Winning Decision Tree: six metrics, all 11 confusion-matrix rows, and classification report.

Supporting local evidence 3

Calibrated RBF SVM: six metrics, 11-class confusion matrix, and classification report.

Captured locally on 2026-11-08. Not BITS Virtual Lab evidence.



Calibrated RBF SVM: six metrics, 11-class confusion matrix, and classification report.

Supporting local evidence 4

Actual feature samples, all 48 column names, and the 11-class test distribution.

Captured locally on 2026-11-08. Not BITS Virtual Lab evidence.

Dataset sample, input schema, and class distribution

Source: UCI Sensorless Drive Diagnosis, 58,509 rows and 48 numeric features. The bundled split has 46,807 training rows and 11,702 test rows (seed 20261106). The final source column is Class; it is excluded from predictors. Feature01 through Feature48 are positional names assigned to unnamed source inputs.

	Class	Feature01	Feature02	Feature03	Feature04	Feature05	Feature06	Feature07	Feature08	Feature09	Feature10	Feature11	Feature12	Feature13	Feature14	Feature15	Feature16	Feature17	Feature18	Feature19	Feature20	Feature21	Feature22	Feature23	Feature24	Feature25	Feature26	Feature27	Feature28	Feature29
0	7	-0.00000003	0.0000004	0.00001	-0.000003	0.000003	0.000008	0.0606	0.0606	0.0606	0.1101	0.1101	0.1101	0.0008	0.0003	0.0006	0.0008	0.0003	0.0006	1.6882	1.6882	1.6882	1.6863	1.6863	1.6863	0.006	-0.2845	0.3845	0.0108	0.7209
1	10	-0.000004	0.00006	0.0002	0.00001	-0.000002	-0.0001	-0.0237	-0.0237	-0.024	-0.0146	-0.0146	-0.0145	0.0022	0.0012	0.0029	0.0022	0.0012	0.0036	1.3352	1.3351	1.3344	1.3352	1.3351	1.3348	-0.0173	0.2369	0.4887	-0.0199	-0.2059
2	2	-0.000006	-0.00002	0.00005	0.0000002	0.00002	0.0002	-0.0282	-0.0282	-0.0283	-0.0354	-0.0354	-0.0356	0.0015	0.0015	0.0028	0.0014	0.0011	0.0024	2.1208	2.1207	2.1203	2.1173	2.1172	2.117	0.0006	0.1908	0.1804	0.0019	0.7876
3	4	-0.000004	-0.00002	-0.00002	0.000005	-0.000004	-0.00005	0.0278	0.0278	0.0278	0.0596	0.0597	0.0597	0.0014	0.0011	0.0017	0.0014	0.0008	0.0024	2.0829	2.0828	2.0826	2.0788	2.0787	2.0784	-0.0031	-0.0002	-0.0972	0.0085	-0.329
4	8	-0.00001	0.000007	0.0003	-0.000009	-0.00005	-0.0002	0.029	0.029	0.0287	0.0311	0.0311	0.0313	0.0026	0.0014	0.0056	0.0027	0.0016	0.0035	1.8786	1.8786	1.8779	1.8727	1.8727	1.8722	0.0004	0.0406	0.6003	0.0248	-0.0495
5	11	-0.00001	0.00008	-0.00006	-0.000003	-0.000001	-0.0001	-0.0808	-0.0809	-0.0808	-0.188	-0.188	-0.1879	0.0014	0.0013	0.002	0.0014	0.0009	0.0025	2.0627	2.0626	2.0623	2.0603	2.0603	2.0599	0.0225	5.3414	-0.5883	0.0041	0.1521
6	2	0.0000004	0.00005	-0.00006	-0.000002	0.00003	0.00009	-0.0312	-0.0313	-0.0312	-0.0449	-0.045	-0.0451	0.0015	0.0009	0.0014	0.0014	0.0008	0.0021	2.1241	2.124	2.1239	2.1211	2.1211	2.1209	-0.0031	1.1215	-0.1515	-0.0075	0.1838
7	8	-0.00002	-0.0000008	-0.00009	-0.000006	-0.000007	-0.0001	0.0145	0.0145	0.0146	0.0186	0.0186	0.0187	0.003	0.0019	0.0033	0.0029	0.0017	0.0042	2.3181	2.3181	2.3176	2.3147	2.3147	2.314	0.0273	-0.2074	0.2268	0.0001	0.0012
8	9	0.000003	-0.000001	-0.00001	0.000005	-0.000006	-0.00002	0.0126	0.0126	0.0126	-0.0162	-0.0162	-0.0161	0.0008	0.0003	0.0006	0.0008	0.0003	0.0005	1.4739	1.4739	1.4739	1.4751	1.4751	1.4751	-0.005	0.0683	0.2826	-0.0156	-0.7861
9	4	0.000005	-0.000003	0.00003	-0.000002	0.0000003	-0.00004	0.017	0.017	0.017	0.0457	0.0457	0.0458	0.0011	0.0004	0.001	0.0011	0.0005	0.0009	1.179	1.179	1.1789	1.1772	1.1772	1.1772	0.0016	-0.054	0.1794	0.0076	0.2982

```
Feature01, Feature02, Feature03, Feature04, Feature05, Feature06, Feature07, Feature08, Feature09,
Feature10, Feature11, Feature12, Feature13, Feature14, Feature15, Feature16, Feature17, Feature18,
Feature19, Feature20, Feature21, Feature22, Feature23, Feature24, Feature25, Feature26, Feature27,
Feature28, Feature29, Feature30, Feature31, Feature32, Feature33, Feature34, Feature35, Feature36,
Feature37, Feature38, Feature39, Feature40, Feature41, Feature42, Feature43, Feature44, Feature45,
Feature46, Feature47, Feature48, Class
```

Condition class	Bundled test rows	Current evaluation rows
1	1064	1064
2	1064	1064
3	1064	1064
4	1064	1064
5	1063	1063
6	1064	1064
7	1064	1064
8	1064	1064
9	1064	1064
10	1064	1064
11	1063	1063

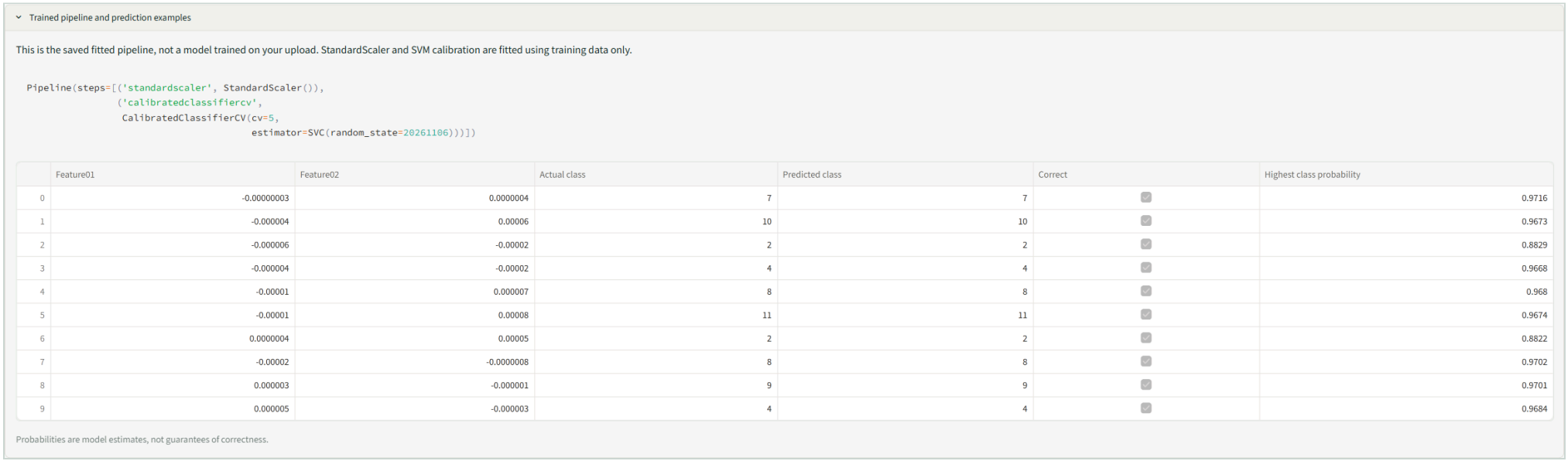
Class codes 1 through 11 are preserved; specific fault-name mappings are not supplied.

Actual feature samples, all 48 column names, and the 11-class test distribution.

Supporting local evidence 5

Saved fitted SVM pipeline and ten actual-versus-predicted condition classes.

Captured locally on 2026-11-08. Not BITS Virtual Lab evidence.



Trained pipeline and prediction examples

This is the saved fitted pipeline, not a model trained on your upload. StandardScaler and SVM calibration are fitted using training data only.

```
Pipeline(steps=[('standardscaler', StandardScaler()),
                ('calibratedclassifiercv',
                 CalibratedClassifierCV(cv=5,
                                        estimator=SVC(random_state=20261106)))])
```

	Feature01	Feature02	Actual class	Predicted class	Correct	Highest class probability
0	-0.00000003	0.0000004	7	7	☑	0.9716
1	-0.000004	0.00006	10	10	☑	0.9673
2	-0.000006	-0.00002	2	2	☑	0.8829
3	-0.000004	-0.00002	4	4	☑	0.9668
4	-0.00001	0.000007	8	8	☑	0.968
5	-0.00001	0.00008	11	11	☑	0.9674
6	0.0000004	0.00005	2	2	☑	0.8822
7	-0.00002	-0.0000008	8	8	☑	0.9702
8	0.000003	-0.000001	9	9	☑	0.9701
9	0.000005	-0.000003	4	4	☑	0.9684

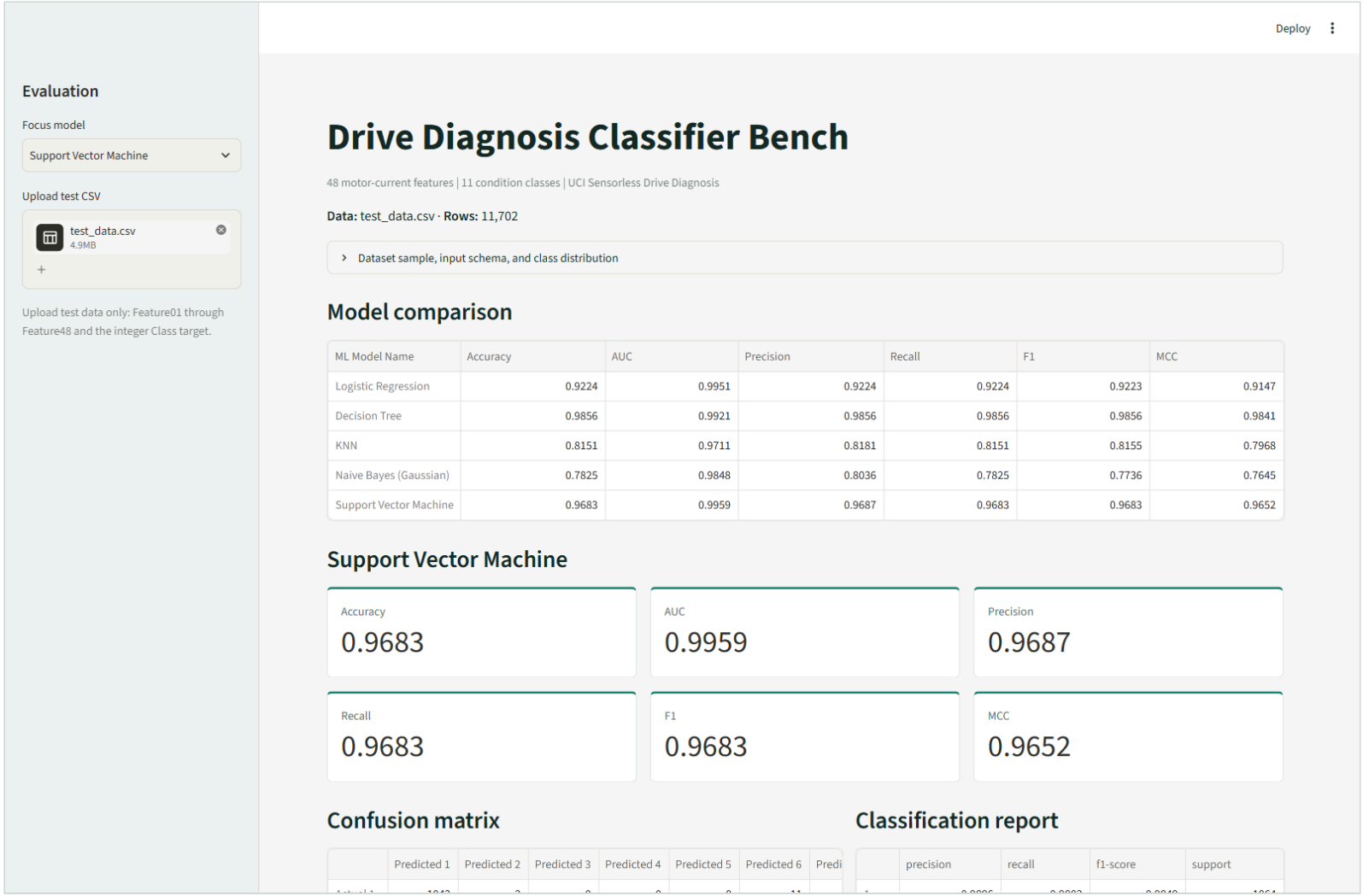
Probabilities are model estimates, not guarantees of correctness.

Saved fitted SVM pipeline and ten actual-versus-predicted condition classes.

Supporting local evidence 6

Successful upload of test_data.csv with all five models evaluated without retraining.

Captured locally on 2026-11-08. Not BITS Virtual Lab evidence.



Successful upload of test_data.csv with all five models evaluated without retraining.

4. README content

Problem Statement

Predict one of 11 drive-condition classes from 48 features extracted from motor current signals. Train the five required classifiers on the same training rows, evaluate six metrics on the same held-out test rows, and inspect predictions in a Streamlit application. This is classification of supplied tabular features, not processing of live motor signals or a deployed safety-monitoring system.

Dataset Description

The UCI Dataset for Sensorless Drive Diagnosis contains 58,509 records. The downloaded file was verified to contain 48 numeric input columns plus one integer target column. It exceeds the assignment minimums of 500 instances and 12 features. UCI reports no missing values; preparation also checks for missing and non-finite values. No exact duplicate full records were found. The full dataset is used, with no sampling or reduction in feature count.

UCI describes intact and defective motor components measured repeatedly under 12 operating conditions involving speed, load moment, and load force. Features were extracted from two-phase current signals using empirical mode decomposition (EMD), intrinsic mode functions and residuals, and statistical summaries such as mean, standard deviation, skewness, and kurtosis.

The source file, `Sensorless_drive_diagnosis.txt`, has no header. Its first 48 columns become `Feature01` through `Feature48`, preserving source order. These are project-assigned positional names, not official physical feature names. The final column becomes `Class`. The target is excluded from model inputs. The available metadata does not establish an exact per-column physical mapping or a reliable fault-name mapping for each class code, so neither is invented.

The exact CSV feature names are:

```
Feature01,Feature02,Feature03,Feature04,Feature05,Feature06,Feature07,Feature08,Feature09,Feature10,Featu
re11,Feature12,Feature13,Feature14,Feature15,Feature16,Feature17,Feature18,Feature19,Feature20,Feature21,
Feature22,Feature23,Feature24,Feature25,Feature26,Feature27,Feature28,Feature29,Feature30,Feature31,Featu
re32,Feature33,Feature34,Feature35,Feature36,Feature37,Feature38,Feature39,Feature40,Feature41,Feature42,
Feature43,Feature44,Feature45,Feature46,Feature47,Feature48
```

Append `Class`, with integer codes from 1 through 11. Preserve column names and source order when preparing upload data.

Split and Class Distribution

A stratified 80/20 split with random state `20261106` produces 46,807 training records and 11,702 test records. Each original class contains exactly 5,319 records; rounding creates a one-record difference in the split counts.

Condition class	Total rows	Training rows	Test rows
1	5,319	4,255	1,064
2	5,319	4,255	1,064
3	5,319	4,255	1,064
4	5,319	4,255	1,064
5	5,319	4,256	1,063
6	5,319	4,255	1,064
7	5,319	4,255	1,064
8	5,319	4,255	1,064
9	5,319	4,255	1,064
10	5,319	4,255	1,064
11	5,319	4,256	1,063
Total	58,509	46,807	11,702

Always predicting one of the largest test classes yields 1,064/11,702 = 0.0909 accuracy. Macro metrics give each condition equal weight; balanced class counts make macro recall close to accuracy, but precision and F1 can still differ.

Example Input Records

The first five actual rows of test_data.csv, with four of the 48 input features shown in scientific notation:

Feature01	Feature02	Feature03	Feature04	Class
-3.9621e-08	4.4180e-07	1.4261e-05	-3.1874e-06	7
-3.5416e-06	6.3153e-05	2.4527e-04	1.2673e-05	10
-5.5443e-06	-1.6878e-05	4.6425e-05	2.6462e-07	2
-3.9820e-06	-2.3080e-05	-2.0704e-05	5.3406e-06	4
-1.3885e-05	7.1854e-06	3.0758e-04	-9.2797e-06	8

Dataset Selection

Sensorless Drive Diagnosis was selected for its engineering interpretation, 48-feature input space, balanced 11-class target, and larger number of records. The following official sources were considered; this is a suitability comparison, not an experiment on those alternatives or a survey of classmates' datasets.

Dataset	Rows	Features	Classes	Consideration
Sensorless Drive Diagnosis	58,509	48 verified	11	Selected: motor-current condition classification with balanced classes.
Steel Plates Faults	1,941	27	7	Smaller industrial dataset; superseded at the user's request.
Image Segmentation	2,310	19	7	Suitable dimensions, but regions originate from only seven images.
Landsat Satellite	6,435	36	6 observed	Suitable dimensions; spatially related neighborhoods and no examples of class 6.

Dataset popularity among other submissions is unknown. Choosing another public dataset does not establish originality. All CSVs, models, metrics, screenshots, and reports in this experiment replace the earlier Steel Plates/Dry Bean work; their scores are not reused.

Citation: Bator, M. (2013). Dataset for Sensorless Drive Diagnosis [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5VP5F. Licensed under CC BY 4.0.

GitHub Repository Link

GitHub Repository

```
MLAssignment2/
|-- app.py
|-- prepare_data.py
|-- train_models.py
|-- test_project.py
|-- requirements.txt
|-- README.md
|-- test_data.csv
|-- model_metrics.csv
|-- model/
    |-- logistic_regression.joblib
    |-- decision_tree.joblib
    |-- knn.joblib
    |-- gaussian_naive_bayes.joblib
    |-- support_vector_machine.joblib
```

The training CSV is generated locally and excluded from Git. Regenerate it before retraining or running the regression tests.

The public repository was accessible during verification on 2026-11-08. Its published README differed from the local Sensorless Drive README. The current local changes still need to be committed and published; an accessible repository link alone does not prove the latest experiment is available there.

Models Used

All five models use 46,807 training rows and 11,702 held-out test rows. Hyperparameters are fixed baselines, not selected by optimizing test scores.

Pipeline Configurations

Model	Preprocessing	Estimator configuration
Logistic Regression	StandardScaler fitted on training rows	C=1.0, L2 regularization, solver=lbfgs, max_iter=2000, random_state=20261106
Decision Tree	None	criterion=gini, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=20261106
KNN	StandardScaler fitted on training rows	n_neighbors=5, weights=uniform, metric=minkowski, p=2
Naive Bayes (Gaussian)	None	GaussianNB, var_smoothing=1e-9
Support Vector Machine	StandardScaler fitted on training rows	SVC with C=1.0, kernel=rbf, gamma=scale, random_state=20261106; CalibratedClassifierCV with method=sigmoid, cv=5

Scaling makes Logistic Regression regularization and KNN/SVM distance calculations less dependent on input magnitudes. Trees split on individual feature thresholds and do not require scaling. SVM probability calibration uses training folds only, not test records; the saved estimator is a calibrated cross-validation pipeline.

Held-Out Results

Precision, recall, and F1 use macro averaging. AUC is macro one-vs-rest AUC from binarized class labels and predicted probabilities. MCC is multiclass Matthews correlation coefficient. model_metrics.csv stores full precision; the report displays four decimal places.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9224	0.9951	0.9224	0.9224	0.9223	0.9147
Decision Tree	0.9856	0.9921	0.9856	0.9856	0.9856	0.9841
KNN	0.8151	0.9711	0.8181	0.8151	0.8155	0.7968
Naive Bayes (Gaussian)	0.7825	0.9848	0.8036	0.7825	0.7736	0.7645
Support Vector Machine	0.9683	0.9959	0.9687	0.9683	0.9683	0.9652

Model Observations

ML Model Name	Observation about model performance
Logistic Regression	Accuracy is 0.9224 and AUC is 0.9951. Linear decision boundaries give a strong baseline but lag Tree and SVM on final class decisions. Standardization keeps regularization from depending primarily on measurement magnitude.
Decision Tree	Leads accuracy, precision, recall, F1 (all approximately 0.9856), and MCC (0.9841). Feature-specific threshold rules work well on this split. The fitted tree has depth 35 and 613 leaves. Its lower AUC than SVM reflects weaker probability ranking despite better final predictions. An unrestricted tree's variance remains a risk, not a demonstrated failure on this test set.
KNN	Accuracy is 0.8151, macro F1 is 0.8155, and AUC is 0.9711. Scaling prevents large-valued features from dominating Euclidean distances, but uniform five-neighbor voting in 48 dimensions can still struggle with redundant dimensions or overlapping neighborhoods. No dimensionality-reduction or neighbor-count ablation was performed.
Naive Bayes (Gaussian)	Lowest accuracy (0.7825), F1 (0.7736), and MCC (0.7645), although AUC is 0.9848. Probability ranking can remain useful while final class decisions are weaker. Current-signal statistical summaries may be dependent and non-Gaussian, challenging its assumptions. Those explanations are hypotheses, not verified causal findings.
Support Vector Machine	Second-highest accuracy (0.9683) and macro F1 (0.9683), and highest AUC (0.9959). Standardized RBF distances allow nonlinear separation, and training-only sigmoid calibration supplies class probabilities. Five-fold calibration increases training and inference costs compared with a single model.
Overall winner for this dataset	Decision Tree on this fixed split, based on the highest label-based metrics and MCC. SVM leads AUC, which evaluates probability ranking rather than final predicted labels. No repeated-split evaluation or significance test establishes universal superiority.

Decision Tree correctly classifies 11,533 of 11,702 test records. Per-class recall ranges from 0.9727 (class 5) to 1.0000 (classes 7 and 11). The high aggregate score is therefore not confined to one dominant class, but perfect class recall on this split does not guarantee performance on future measurements.

The record-level random split does not demonstrate generalization to a new motor, unseen operating condition, or independent measurement session. UCI reports repeated measurements under multiple operating conditions, but the published table does not provide a verified session/group identifier for a grouped split. High held-out scores must be interpreted within that limitation.

Experiment Workflow

1. Install pinned dependencies in Python 3.11.
2. Run prepare_data.py to download the full UCI source, validate 49 columns, assign positional feature names, and make a stratified training/test split.
3. Run train_models.py to fit all five pipelines on training rows only. Scaling and SVM calibration never use held-out test rows.
4. Calculate six test metrics and save fitted pipelines and full-precision results.
5. Run test_project.py to validate source parsing, labels, split separation, scalers, saved metrics, and all five UI model selections.
6. Start app.py and inspect sample data, class counts, the comparison, selected model metrics, confusion matrix, classification report, and predictions.
7. Upload labeled test data only. The app performs inference without retraining. Comparisons are cached by input data and model modification times so selecting a different model does not repeat full-dataset inference.

8. Publish the replacement artifacts, redeploy, verify both public links, and assemble the final PDF with genuine BITS Virtual Lab evidence if available.

Reproduce the Local Run

```
uv venv .venv --python 3.11
uv pip install --python .venv\Scripts\python.exe -r requirements.txt
.venv\Scripts\python.exe prepare_data.py
.venv\Scripts\python.exe train_models.py
.venv\Scripts\python.exe -m unittest test_project -v
.venv\Scripts\python.exe -m streamlit run app.py
```

Training on the full dataset, particularly five-fold calibrated SVM, takes longer than the earlier smaller datasets. The first app evaluation also performs inference for all five models; subsequent focus changes reuse cached comparisons. The latest verification on 2026-11-08 completed all six regression tests in approximately 161 seconds, including one full-data app comparison. Earlier runs took approximately 61 and 334 seconds, demonstrating that elapsed time varies with machine load and cache state. Public cloud runtime and memory limits still need verification; local timing does not certify acceptable cloud response time.

Use pinned dependencies, including `scikit-learn==1.9.1`, to reproduce the saved models. Another scikit-learn version may warn or fail to reproduce behavior.

For BITS Virtual Lab, install the same dependencies and run preparation, training, and the app there. If downloads are blocked, transfer the generated CSVs and run training in the lab. Capture genuine lab execution evidence.

Streamlit App

The intended public link is Drive Diagnosis Classifier Bench. During verification on 2026-11-08, the browser was redirected to a network "Site Blocked" page; a separate fetch returned HTTP 403. This prevents verification from this environment and does not establish whether the deployed application itself is unavailable. Local artifacts are not published automatically. Commit/push, redeploy, and check the URL from an allowed network before submission.

The app provides:

- Test CSV upload with 48 numeric features and the integer `Class` target.
- Five-model comparison, focus-model selection, and six evaluation metrics.
- An 11-class confusion matrix and classification report.
- Actual input samples, exact schema, and evaluation class distribution.
- The fitted pipeline and ten actual-versus-predicted examples with estimated maximum class probabilities, which are not guarantees of correctness.

All condition classes 1 through 11 must be present in the evaluation CSV. Missing required columns, missing/non-finite values, nonnumeric inputs, and unknown/noninteger classes are rejected explicitly.

Submission Checklist

The final PDF must contain:

1. GitHub repository link with source, dependencies, README, saved models, and experiment test CSV.

2. A working live Streamlit app link.
3. One genuine BITS Virtual Lab execution screenshot.
4. The GitHub README content.

> Important: Local screenshots are not BITS Virtual Lab evidence. Without genuine lab evidence, the backup does not meet the 1-mark execution requirement. An inaccessible public app may affect additional deployment-related marks.

Review and understand the implementation before submission. The assignment permits AI for learning support, not direct copy-paste submissions. Retain genuine commit history; a dataset change does not prove originality.

Guideline Verification

These checks apply to the current local Sensorless Drive experiment. They are not a guarantee of marks or a certification of originality.

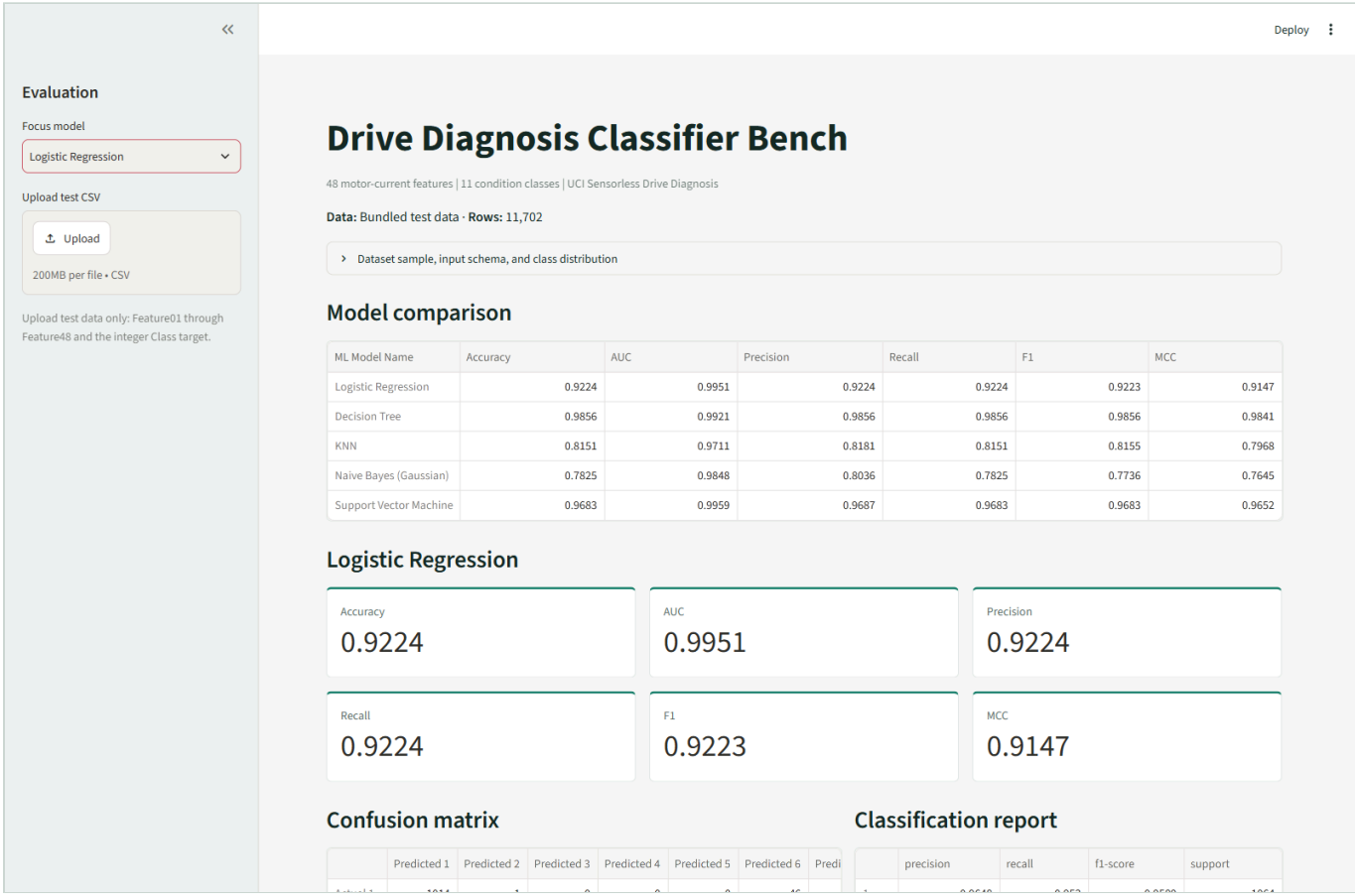
Assignment requirement	Verification result
One public classification dataset, at least 500 instances and 12 features	Verified: UCI Sensorless Drive Diagnosis, 58,509 records, 48 inputs, and 11 classes.
Five required classifiers on the same dataset	Verified: Logistic Regression, Decision Tree, KNN, Gaussian Naive Bayes, and RBF SVM use the same 46,807 training and 11,702 test records.
Six metrics for every classifier	Verified: all 30 saved values match app calculations to a tolerance of 1e-12; the README table matches their four-decimal rounding.
Observations for every model and an overall winner	Included: five analytical observations, Decision Tree winner, and evaluation limitations.
Source, dependencies, test CSV, and five saved models	Present locally. Public repository is accessible, but the current local revision has not been published.
Test CSV upload without retraining	Verified in the browser using the bundled test CSV; missing-feature input is rejected explicitly. Model files are unchanged by inference.
Model dropdown, comparison, six metrics, confusion matrix/report	Verified for all five choices in regression tests and shown in fresh local screenshots.
Single PDF ordered as repository link, live app link, BITS evidence, README	Backup uses these section positions; the BITS section states that genuine lab evidence is unavailable. Word is an editing copy, not the final submission format.
Live Streamlit app opens an interactive frontend	Not verified: this environment blocks the public URL. Check the redeployed app from an allowed network.
One genuine BITS Virtual Lab screenshot	Not fulfilled: all included images are explicitly local evidence. Replace the evidence section after a genuine lab run.
Academic integrity and genuine history	Must be reviewed by the author; tests and a dataset change cannot establish originality or compliance with the AI-use policy.

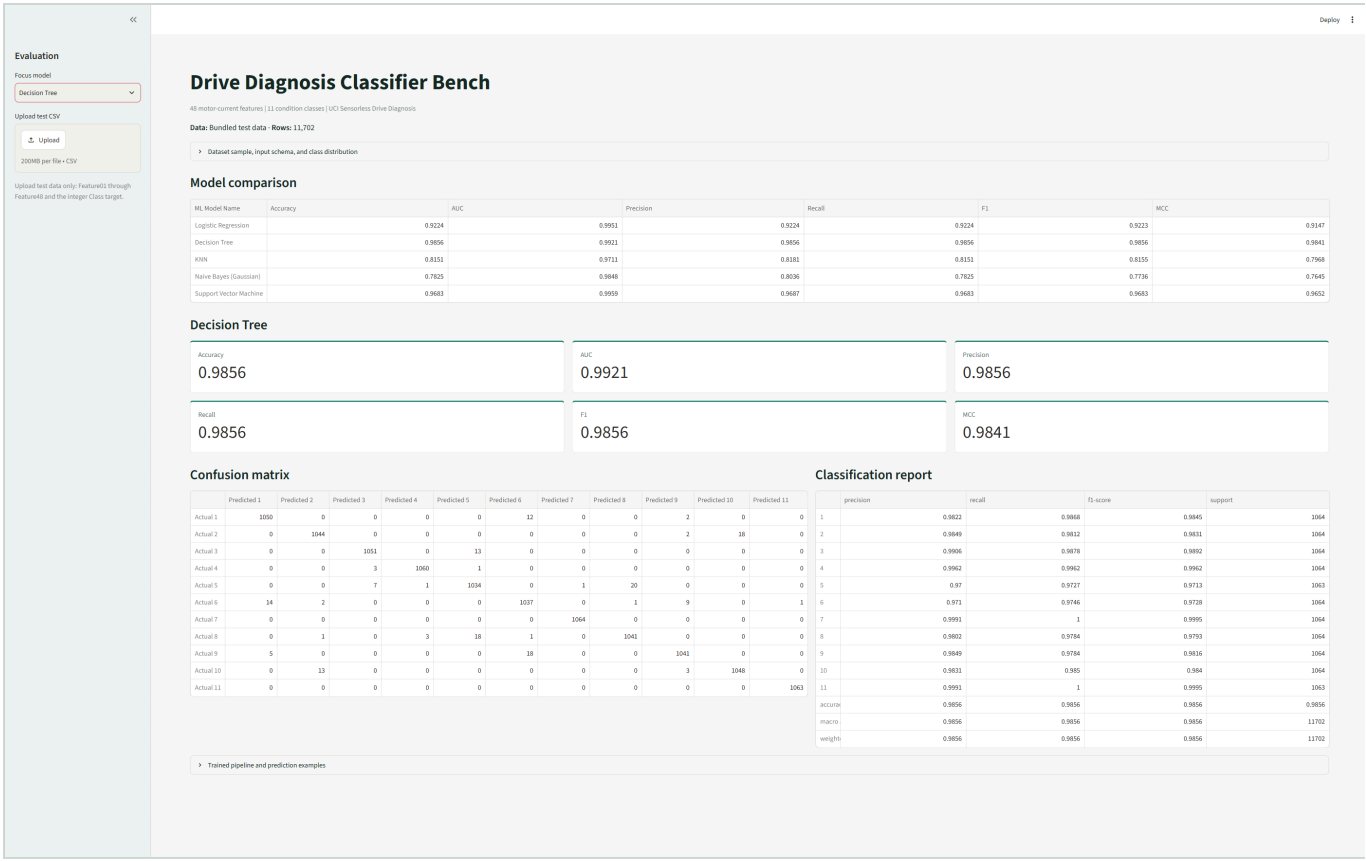
Before final submission, publish the current artifacts, verify the live app, replace the evidence section if a BITS run succeeds, and export the edited Word document as a single PDF. Recheck the PDF links and tables after any manual edits.

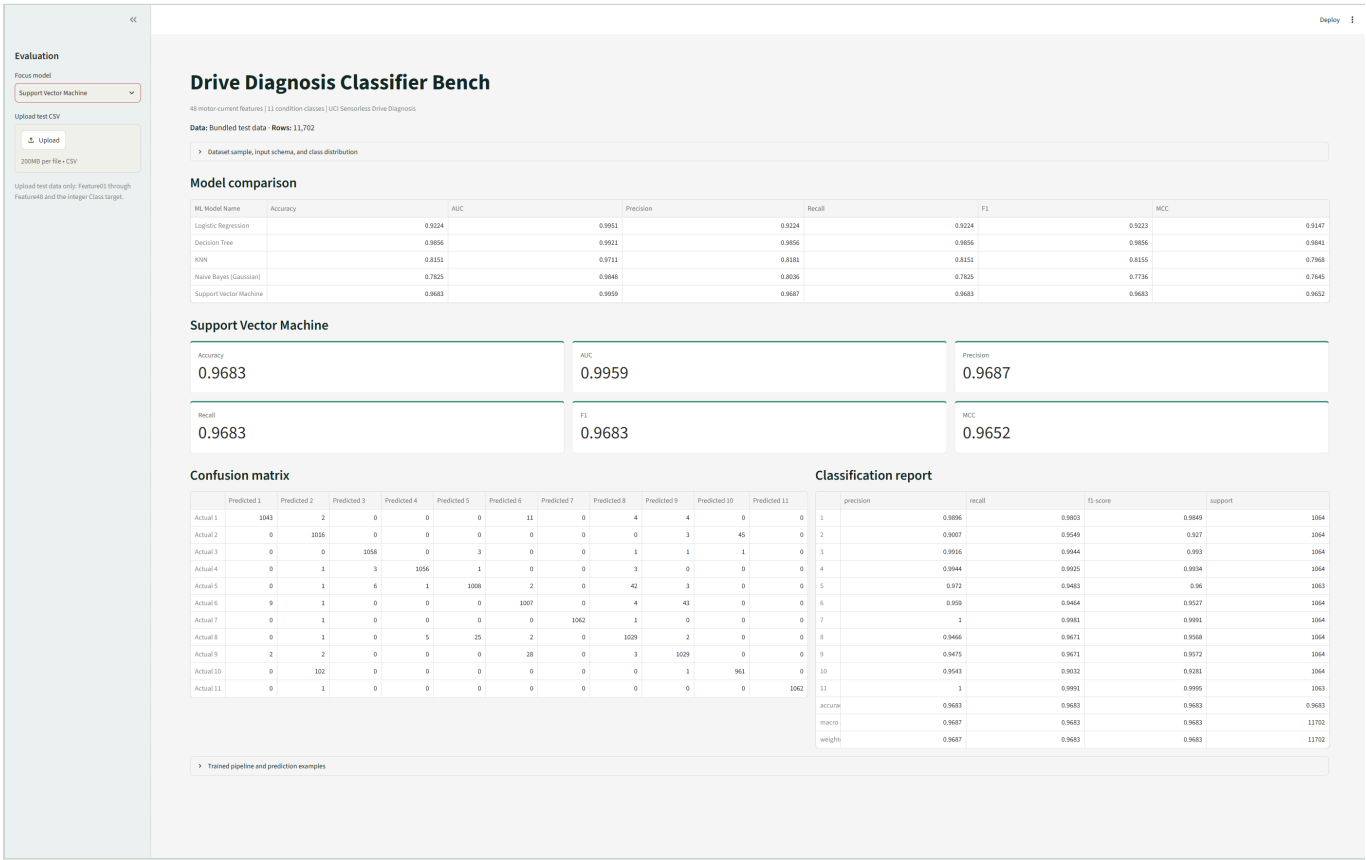
Local Backup Evidence

The backup PDF and editable Word backup contain the current experiment. Replace the local execution section with genuine BITS evidence if a run succeeds.

All screenshots below were refreshed from genuine local execution on 2026-11-08. They show the same Sensorless Drive test split and models as the result table.




Deploy
Evaluation
Focus model
Decision Tree
Upload test CSV
Upload
200MB per file • CSV
Upload test data only: Feature01 through Feature48 and the integer Class target.
Drive Diagnosis Classifier Bench
48 motor-current features | 11 condition classes | UCI Sensorless Drive Diagnosis
Data: Bundled test data · Rows: 11,702
Dataset sample, input schema, and class distribution
Model comparison
ML Model Name | Accuracy | AUC | Precision | Recall | F1 | MCC
Logistic Regression | 0.9224 | 0.9951 | 0.9224 | 0.9224 | 0.9223 | 0.9147
Decision Tree | 0.9856 | 0.9921 | 0.9856 | 0.9856 | 0.9856 | 0.9841
KNN | 0.8151 | 0.9711 | 0.8181 | 0.8151 | 0.8155 | 0.7968
Naive Bayes (Gaussian) | 0.7825 | 0.9848 | 0.8036 | 0.7825 | 0.7736 | 0.7645
Support Vector Machine | 0.9683 | 0.9959 | 0.9687 | 0.9683 | 0.9683 | 0.9652
Decision Tree
Accuracy 0.9856
AUC 0.9921
Precision 0.9856
Recall 0.9856
F1 0.9856
MCC 0.9841
Confusion matrix
Predicted 1 | Predicted 2 | Predicted 3 | Predicted 4 | Predicted 5 | Predicted 6 | Predicted 7 | Predicted 8 | Predicted 9 | Predicted 10 | Predicted 11
Actual 1 | 1050 | 0 | 0 | 0 | 0 | 12 | 0 | 0 | 2 | 0 | 0
Actual 2 | 0 | 1044 | 0 | 0 | 0 | 0 | 0 | 0 | 2 | 18 | 0
Actual 3 | 0 | 0 | 1051 | 0 | 13 | 0 | 0 | 0 | 0 | 0 | 0
Actual 4 | 0 | 0 | 3 | 1060 | 1 | 0 | 0 | 0 | 0 | 0 | 0
Actual 5 | 0 | 0 | 7 | 1 | 1034 | 0 | 1 | 20 | 0 | 0 | 0
Actual 6 | 14 | 2 | 0 | 0 | 0 | 1037 | 0 | 1 | 9 | 0 | 1
Actual 7 | 0 | 0 | 0 | 0 | 0 | 0 | 1064 | 0 | 0 | 0 | 0
Actual 8 | 0 | 1 | 0 | 3 | 18 | 1 | 0 | 1041 | 0 | 0 | 0
Actual 9 | 5 | 0 | 0 | 0 | 0 | 18 | 0 | 0 | 1041 | 0 | 0
Actual 10 | 0 | 13 | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 1048 | 0
Actual 11 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1063
Classification report
precision | recall | f1-score | support
1 | 0.9822 | 0.9868 | 0.9845 | 1064
2 | 0.9849 | 0.9812 | 0.9831 | 1064
3 | 0.9906 | 0.9878 | 0.9892 | 1064
4 | 0.9962 | 0.9962 | 0.9962 | 1064
5 | 0.97 | 0.9727 | 0.9713 | 1063
6 | 0.971 | 0.9746 | 0.9728 | 1064
7 | 0.9991 | 1 | 0.9995 | 1064
8 | 0.9802 | 0.9784 | 0.9793 | 1064
9 | 0.9849 | 0.9784 | 0.9816 | 1064
10 | 0.9831 | 0.985 | 0.984 | 1064
11 | 0.9991 | 1 | 0.9995 | 1063
accurac | 0.9856 | 0.9856 | 0.9856 | 0.9856
macro | 0.9856 | 0.9856 | 0.9856 | 11702
weighti | 0.9856 | 0.9856 | 0.9856 | 11702
Trained pipeline and prediction examples


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KNN | 0.8151 | 0.9711 | 0.8181 | 0.8151 | 0.8155 | 0.7968
Naive Bayes (Gaussian) | 0.7825 | 0.9848 | 0.8036 | 0.7825 | 0.7736 | 0.7645
Support Vector Machine | 0.9683 | 0.9959 | 0.9687 | 0.9683 | 0.9683 | 0.9652
Support Vector Machine
Accuracy 0.9683
AUC 0.9959
Precision 0.9687
Recall 0.9683
F1 0.9683
MCC 0.9652
Confusion matrix
Predicted 1 | Predicted 2 | Predicted 3 | Predicted 4 | Predicted 5 | Predicted 6 | Predicted 7 | Predicted 8 | Predicted 9 | Predicted 10 | Predicted 11
Actual 1 | 1043 | 2 | 0 | 0 | 0 | 11 | 0 | 4 | 4 | 0 | 0
Actual 2 | 0 | 1016 | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 45 | 0
Actual 3 | 0 | 0 | 1058 | 0 | 3 | 0 | 0 | 1 | 1 | 1 | 0
Actual 4 | 0 | 1 | 3 | 1056 | 1 | 0 | 0 | 3 | 0 | 0 | 0
Actual 5 | 0 | 1 | 6 | 1 | 1008 | 2 | 0 | 42 | 3 | 0 | 0
Actual 6 | 9 | 1 | 0 | 0 | 0 | 1007 | 0 | 4 | 43 | 0 | 0
Actual 7 | 0 | 1 | 0 | 0 | 0 | 0 | 1062 | 1 | 0 | 0 | 0
Actual 8 | 0 | 1 | 0 | 5 | 25 | 2 | 0 | 1029 | 2 | 0 | 0
Actual 9 | 2 | 2 | 0 | 0 | 0 | 28 | 0 | 3 | 1029 | 0 | 0
Actual 10 | 0 | 102 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 961 | 0
Actual 11 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1062
Classification report
precision | recall | f1-score | support
1 | 0.9896 | 0.9803 | 0.9849 | 1064
2 | 0.9007 | 0.9549 | 0.927 | 1064
3 | 0.9916 | 0.9944 | 0.993 | 1064
4 | 0.9944 | 0.9925 | 0.9934 | 1064
5 | 0.972 | 0.9483 | 0.96 | 1063
6 | 0.959 | 0.9464 | 0.9527 | 1064
7 | 1 | 0.9981 | 0.9991 | 1064
8 | 0.9466 | 0.9671 | 0.9568 | 1064
9 | 0.9475 | 0.9671 | 0.9572 | 1064
10 | 0.9543 | 0.9032 | 0.9281 | 1064
11 | 1 | 0.9991 | 0.9995 | 1063
accurac | 0.9683 | 0.9683 | 0.9683 | 0.9683
macro | 0.9687 | 0.9683 | 0.9683 | 11702
weighti | 0.9687 | 0.9683 | 0.9683 | 11702
Trained pipeline and prediction examples

Dataset sample, input schema, and class distribution

Source: UCI Sensorless Drive Diagnosis, 58,509 rows and 48 numeric features. The bundled split has 46,807 training rows and 11,702 test rows (seed 20261106). The final source column is Class; it is excluded from predictors. Feature01 through Feature48 are positional names assigned to unnamed source inputs.

	Class	Feature01	Feature02	Feature03	Feature04	Feature05	Feature06	Feature07	Feature08	Feature09	Feature10	Feature11	Feature12	Feature13	Feature14	Feature15	Feature16	Feature17	Feature18	Feature19	Feature20	Feature21	Feature22	Feature23	Feature24	Feature25	Feature26	Feature27	Feature28	Feature29
0	7	-0.00000003	0.0000004	0.00001	-0.000003	0.000003	0.000008	0.0606	0.0606	0.0606	0.1101	0.1101	0.1101	0.0008	0.0003	0.0006	0.0008	0.0003	0.0006	1.6882	1.6882	1.6882	1.6863	1.6863	1.6863	0.006	-0.2845	0.3845	0.0108	0.7209
1	10	-0.000004	0.00006	0.0002	0.00001	-0.000002	-0.0001	-0.0237	-0.0237	-0.024	-0.0146	-0.0146	-0.0145	0.0022	0.0012	0.0029	0.0022	0.0012	0.0036	1.3352	1.3351	1.3344	1.3352	1.3351	1.3348	-0.0173	0.2369	0.4887	-0.0199	-0.2059
2	2	-0.000006	-0.00002	0.00005	0.0000002	0.00002	0.0002	-0.0282	-0.0282	-0.0283	-0.0354	-0.0354	-0.0356	0.0015	0.0015	0.0028	0.0014	0.0011	0.0024	2.1208	2.1207	2.1203	2.1173	2.1172	2.117	0.0006	0.1908	0.1804	0.0019	0.7876
3	4	-0.000004	-0.00002	-0.00002	0.000005	-0.000004	-0.00005	0.0278	0.0278	0.0278	0.0596	0.0597	0.0597	0.0014	0.0011	0.0017	0.0014	0.0008	0.0024	2.0829	2.0828	2.0826	2.0788	2.0787	2.0784	-0.0031	-0.0002	-0.0972	0.0085	-0.329
4	8	-0.00001	0.000007	0.0003	-0.000009	-0.00005	-0.0002	0.029	0.029	0.0287	0.0311	0.0311	0.0313	0.0026	0.0014	0.0056	0.0027	0.0016	0.0035	1.8786	1.8786	1.8779	1.8727	1.8727	1.8722	0.0004	0.0406	0.6003	0.0248	-0.0495
5	11	-0.00001	0.00008	-0.00006	-0.000003	-0.000001	-0.0001	-0.0808	-0.0809	-0.0808	-0.188	-0.188	-0.1879	0.0014	0.0013	0.002	0.0014	0.0009	0.0025	2.0627	2.0626	2.0623	2.0603	2.0603	2.0599	0.0225	5.3414	-0.5883	0.0041	0.1521
6	2	0.0000004	0.00005	-0.00006	-0.000002	0.00003	0.00009	-0.0312	-0.0313	-0.0312	-0.0449	-0.045	-0.0451	0.0015	0.0009	0.0014	0.0014	0.0008	0.0021	2.1241	2.124	2.1239	2.1211	2.1211	2.1209	-0.0031	1.1215	-0.1515	-0.0075	0.1838
7	8	-0.00002	-0.0000008	-0.00009	-0.000006	-0.000007	-0.0001	0.0145	0.0145	0.0146	0.0186	0.0186	0.0187	0.003	0.0019	0.0033	0.0029	0.0017	0.0042	2.3181	2.3181	2.3176	2.3147	2.3147	2.314	0.0273	-0.2074	0.2268	0.0001	0.0012
8	9	0.000003	-0.000001	-0.00001	0.000005	-0.000006	-0.00002	0.0126	0.0126	0.0126	-0.0162	-0.0162	-0.0161	0.0008	0.0003	0.0006	0.0008	0.0003	0.0005	1.4739	1.4739	1.4739	1.4751	1.4751	1.4751	-0.005	0.0683	0.2826	-0.0156	-0.7861
9	4	0.000005	-0.000003	0.00003	-0.000002	0.0000003	-0.00004	0.017	0.017	0.017	0.0457	0.0457	0.0458	0.0011	0.0004	0.001	0.0011	0.0005	0.0009	1.179	1.179	1.1789	1.1772	1.1772	1.1772	0.0016	-0.054	0.1794	0.0076	0.2982

```
Feature01, Feature02, Feature03, Feature04, Feature05, Feature06, Feature07, Feature08, Feature09,
Feature10, Feature11, Feature12, Feature13, Feature14, Feature15, Feature16, Feature17, Feature18,
Feature19, Feature20, Feature21, Feature22, Feature23, Feature24, Feature25, Feature26, Feature27,
Feature28, Feature29, Feature30, Feature31, Feature32, Feature33, Feature34, Feature35, Feature36,
Feature37, Feature38, Feature39, Feature40, Feature41, Feature42, Feature43, Feature44, Feature45,
Feature46, Feature47, Feature48, Class
```

Condition class	Bundled test rows	Current evaluation rows
1	1064	1064
2	1064	1064
3	1064	1064
4	1064	1064
5	1063	1063
6	1064	1064
7	1064	1064
8	1064	1064
9	1064	1064
10	1064	1064
11	1063	1063

Class codes 1 through 11 are preserved; specific fault-name mappings are not supplied.

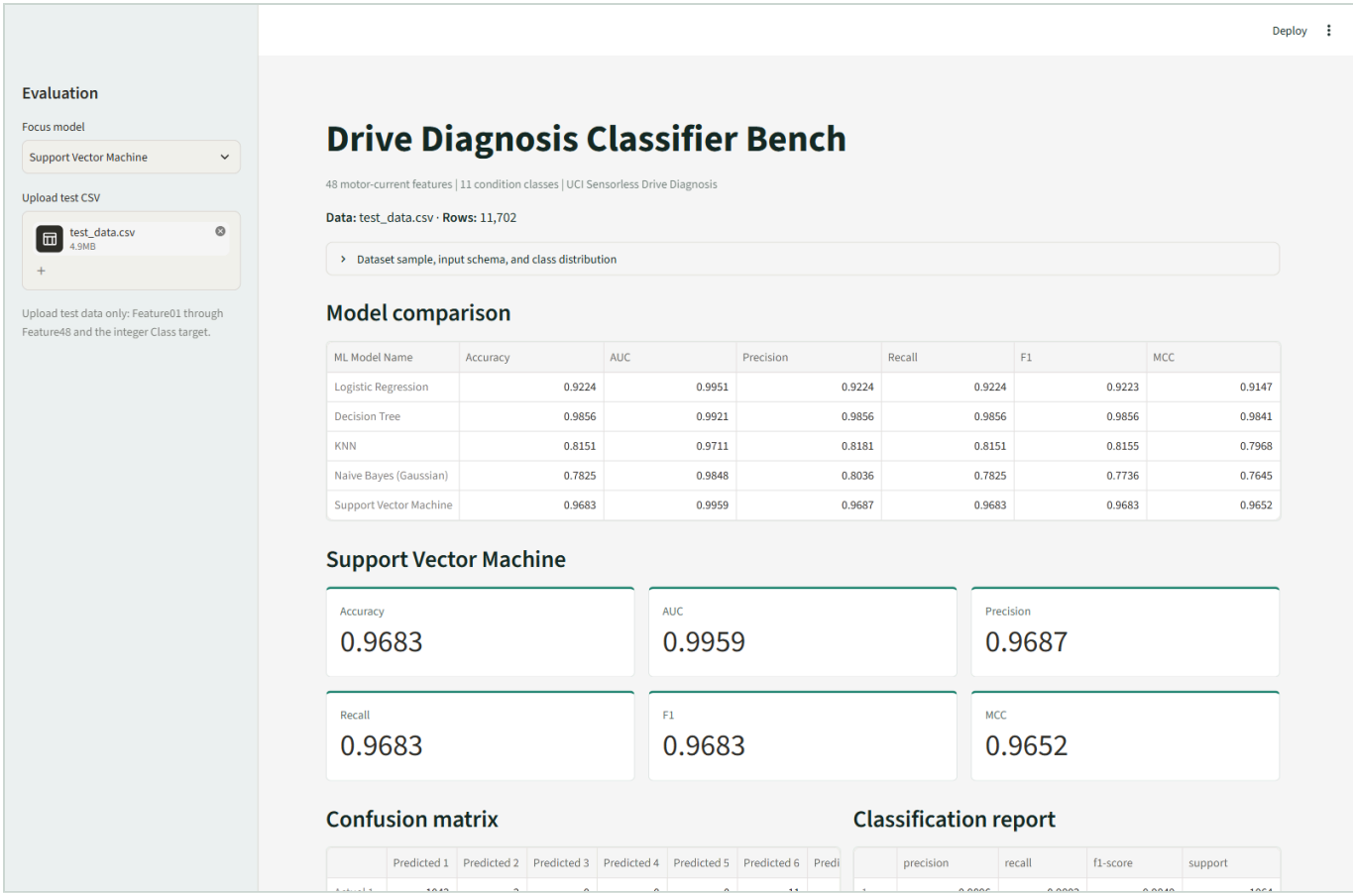
Trained pipeline and prediction examples

This is the saved fitted pipeline, not a model trained on your upload. StandardScaler and SVM calibration are fitted using training data only.

```
Pipeline(steps=[('standardscaler', StandardScaler()),
                ('calibratedclassifiercv',
                 CalibratedClassifierCV(cv=5,
                                        estimator=SVC(random_state=20261106)))])
```

	Feature01	Feature02	Actual class	Predicted class	Correct	Highest class probability
0	-0.00000003	0.0000004	7	7	☑	0.9716
1	-0.000004	0.00006	10	10	☑	0.9673
2	-0.000006	-0.00002	2	2	☑	0.8829
3	-0.000004	-0.00002	4	4	☑	0.9668
4	-0.00001	0.000007	8	8	☑	0.968
5	-0.00001	0.00008	11	11	☑	0.9674
6	0.0000004	0.00005	2	2	☑	0.8822
7	-0.00002	-0.0000008	8	8	☑	0.9702
8	0.000003	-0.000001	9	9	☑	0.9701
9	0.000005	-0.000003	4	4	☑	0.9684

Probabilities are model estimates, not guarantees of correctness.


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4.9MB
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Support Vector Machine
Accuracy 0.9683
AUC 0.9959
Precision 0.9687
Recall 0.9683
F1 0.9683
MCC 0.9652
Confusion matrix
Predicted 1 | Predicted 2 | Predicted 3 | Predicted 4 | Predicted 5 | Predicted 6 | Predi
Classification report
precision | recall | f1-score | support